

# GraphTheorySymbol:

## A set of $\text{\LaTeX}$ commands to have symbols designed for Graph Theory

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### Abstract

This documentation is for  $\text{\LaTeX}$  package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

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## 1 Introduction

This documentation is for  $\text{\LaTeX}$  package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com:LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first  $\text{\LaTeX}$  package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better  $\text{\LaTeX}$  code, please submit a pull request on GitHub.

## 2 Usage

### 2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

| Long name                                                        | Short name               | Result  | In context  |
|------------------------------------------------------------------|--------------------------|---------|-------------|
| <code>\GTSadjacency/</code>                                      | <code>\GTSa/</code>      | $\circ$ | $u \circ v$ |
| <code>\GTSadjacencyRelation/</code>                              | <code>\GTSaR/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacency/</code>                                    | <code>\GTSu/</code>      | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacencyRelation/</code>                            | <code>\GTSuR/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUp/</code>                            | <code>\GTSdAU/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUpRelation/</code>                    | <code>\GTSdAUR/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUp/</code>                     | <code>\GTSnDAU/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUpRelation/</code>             | <code>\GTSnDAUR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDown/</code>                          | <code>\GTSdAD/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDownRelation/</code>                  | <code>\GTSdADR/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDown/</code>                   | <code>\GTSnDAD/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDownRelation/</code>           | <code>\GTSnDADR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSincidence/</code>                                      | <code>\GTSi/</code>      | $\circ$ | $v \circ e$ |
| <code>\GTSincidenceRelation/</code>                              | <code>\GTSiR/</code>     | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidence/</code>                               | <code>\GTSnI/</code>     | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidenceRelation/</code>                       | <code>\GTSnIR/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidence/</code>                                   | <code>\GTSoI/</code>     | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidenceRelation/</code>                           | <code>\GTSoIR/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidence/</code>                            | <code>\GTSnOI/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidenceRelation/</code>                    | <code>\GTSnOIR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidence/</code>                                    | <code>\GTSiI/</code>     | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidenceRelation/</code>                            | <code>\GTSiIR/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidence/</code>                             | <code>\GTSnII/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidenceRelation/</code>                     | <code>\GTSnIIR/</code>   | $\circ$ | $v \circ e$ |
| Long name                                                        | Short name               | Result  | In context  |
| <code>\GTSadjacencyMonochrome/</code>                            | <code>\GTSaM/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSadjacencyMonochromeRelation/</code>                    | <code>\GTSaMR/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacencyMonochrome/</code>                          | <code>\GTSuM/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacencyMonochromeRelation/</code>                  | <code>\GTSuMR/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUpMonochrome/</code>                  | <code>\GTSdAUM/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>          | <code>\GTSdAUMR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>           | <code>\GTSnDAUM/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>   | <code>\GTSnDAUMR/</code> | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDownMonochrome/</code>                | <code>\GTSdADM/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>        | <code>\GTSdADMR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>         | <code>\GTSnDADM/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code> | <code>\GTSnDADMR/</code> | $\circ$ | $u \circ v$ |
| <code>\GTSincidenceMonochrome/</code>                            | <code>\GTSiM/</code>     | $\circ$ | $v \circ e$ |
| <code>\GTSincidenceMonochromeRelation/</code>                    | <code>\GTSiMR/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidenceMonochrome/</code>                     | <code>\GTSnIM/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidenceMonochromeRelation/</code>             | <code>\GTSnIMR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidenceMonochrome/</code>                         | <code>\GTSoIM/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidenceMonochromeRelation/</code>                 | <code>\GTSoIMR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidenceMonochrome/</code>                  | <code>\GTSnOIM/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidenceMonochromeRelation/</code>          | <code>\GTSnOIMR/</code>  | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidenceMonochrome/</code>                          | <code>\GTSiIM/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidenceMonochromeRelation/</code>                  | <code>\GTSiIMR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidenceMonochrome/</code>                   | <code>\GTSnIIM/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidenceMonochromeRelation/</code>           | <code>\GTSnIIMR/</code>  | $\circ$ | $v \circ e$ |

|                                           |            |        |            |
|-------------------------------------------|------------|--------|------------|
| \GTSsetMainColor{blue}                    |            |        |            |
| Long name                                 | Short name | Result | In context |
| \GTSadjacency/                            | \GTSa/     |        | u  v       |
| \GTSadjacencyRelation/                    | \GTSaR/    |        | u  v       |
| \GTSunadjacency/                          | \GTSu/     |        | u  v       |
| \GTSunadjacencyRelation/                  | \GTSuR/    |        | u  v       |
| \GTSdirectedAdjacencyUp/                  | \GTSdAU/   |        | u  v       |
| \GTSdirectedAdjacencyUpRelation/          | \GTSdAUR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUp/           | \GTSnDAU/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUpRelation/   | \GTSnDAUR/ |        | u  v       |
| \GTSdirectedAdjacencyDown/                | \GTSdAD/   |        | u  v       |
| \GTSdirectedAdjacencyDownRelation/        | \GTSdADR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDown/         | \GTSnDAD/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDownRelation/ | \GTSnDADR/ |        | u  v       |
| \GTSincidence/                            | \GTSi/     |        | v  e       |
| \GTSincidenceRelation/                    | \GTSiR/    |        | v  e       |
| \GTSnegatedIncidence/                     | \GTSnI/    |        | v  e       |
| \GTSnegatedIncidenceRelation/             | \GTSnIR/   |        | v  e       |
| \GTSoutIncidence/                         | \GTSOI/    |        | v  e       |
| \GTSoutIncidenceRelation/                 | \GTSOIR/   |        | v  e       |
| \GTSnegatedOutIncidence/                  | \GTSnOI/   |        | v  e       |
| \GTSnegatedOutIncidenceRelation/          | \GTSnOIR/  |        | v  e       |
| \GTSinIncidence/                          | \GTSiI/    |        | v  e       |
| \GTSinIncidenceRelation/                  | \GTSiIR/   |        | v  e       |
| \GTSnegatedInIncidence/                   | \GTSnII/   |        | v  e       |
| \GTSnegatedInIncidenceRelation/           | \GTSnIIR/  |        | v  e       |

## 2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`\{<commandname>\}` The name of the command to create.

`\{<commandcode>\}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`\{<color>\}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`\{<color>\}` The color that must be used by GraphTheorySymbol afterward.

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`\{<mainColor>\}` If non-empty, this color will be used as main color.

`\{<secondColor>\}` If non-empty, this color will be used as second color.

`\{<drawSlash>\}` Non-empty implies that a negation will be drawn.

`\{<drawUpArrow>\}` Non-empty implies that an up-arrow will be drawn.

`\{<drawDownArrow>\}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`\{<mainColor>\}` If non-empty, this color will be used as main color.

`\{<secondColor>\}` If non-empty, this color will be used as second color.

`\{<drawSlash>\}` Non-empty implies that a negation will be drawn.

`\{<drawUpArrow>\}` Non-empty implies that an up-arrow will be drawn.

`\{<drawDownArrow>\}` Non-empty implies that a down-arrow will be drawn.

`\{<relation>\}` Non-empty implies that relation spacing will be added.

`\GTSxa` Versatile macros for adjacency that can be given options.

`\GTSxadj` Argument:

`\GTSxadjacency` [`<optionsList>`] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,

secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,

monochrome or mono or mon or m (equivalent to mC=black,sC=black),

negated or not or n,

up or u,

down or d,

relation or rel or r.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small

`\GTSadj` dots in the middle of the two circles corresponding to the graph vertices. But it was too

`\GTSa` similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as  $\text{Adj}(.,.)$ .

`A.\GTSadjacency/A.` or `A.\GTSadj/A.` or `A.\GTSa/A.` will yield  $A.\mathring{A}$ .

`u \GTSadjacency/ v` or `u \GTSadj/ v` or `u \GTSa/ v` will yield  $u \mathring{v}$ .

`\$u \GTSadjacency/ v\$` or `\$u \GTSadj/ v\$` or `\$u \GTSa/ v\$` will yield  $u \mathring{v}$ .

`\GTSadjacencyRelation` This macro will draw an adjacency symbol.

`\GTSadjRel` `\$u\GTSadjacencyRelation/v\$` or `\$u\GTSadjRel/v\$` or `\$u\GTSaR/v\$` will yield  $u \mathring{v}$ .

`\GTSaR`

`\GTSunadjacency` This macro will draw an unadjacency symbol.

`\GTSunadj` `A.\GTSunadjacency/A.` or `A.\GTSunadj/A.` or `A.\GTSu/A.` will yield  $A.\mathring{A}$ .

`\GTSu`

$u \setminus \text{GTSunadjacency} / v$  or  $u \setminus \text{GTSunadj} / v$  or  $u \setminus \text{GTSu} / v$  will yield  $u \setminus v$ .  
 $\$u \setminus \text{GTSunadjacency} / v\$$  or  $\$u \setminus \text{GTSunadj} / v\$$  or  $\$u \setminus \text{GTSu} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSunadjacencyRelation}$  This macro will draw an unadjacency symbol.  
 $\setminus \text{GTSunadjRel}$   $\$u \setminus \text{GTSunadjacencyRelation} / v\$$  or  $\$u \setminus \text{GTSunadjRel} / v\$$  or  $\$u \setminus \text{GTSuR} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSuR}$   
 $\setminus \text{GTSdirectedAdjacencyUp}$  This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.  
 $\setminus \text{GTSdirAdjUp}$   $A. \setminus \text{GTSdirectedAdjacencyUp} / A.$  or  $A. \setminus \text{GTSdirAdjUp} / A.$  or  $A. \setminus \text{GTSdAU} / A.$  will yield  $A. \setminus A.$   
 $\setminus \text{GTSdAU}$   $u \setminus \text{GTSdirectedAdjacencyUp} / v$  or  $u \setminus \text{GTSdirAdjUp} / v$  or  $u \setminus \text{GTSdAU} / v$  will yield  $u \setminus v$ .  
 $\$u \setminus \text{GTSdirectedAdjacencyUp} / v\$$  or  $\$u \setminus \text{GTSdirAdjUp} / v\$$  or  $\$u \setminus \text{GTSdAU} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSdirectedAdjacencyUpRelation}$  This macro will draw a directed adjacency upward symbol with relation spacing in math mode.  
 $\setminus \text{GTSdirAdjUpRel}$   $\$u \setminus \text{GTSdirectedAdjacencyUpRelation} / v\$$  or  $\$u \setminus \text{GTSdirAdjUpRel} / v\$$  or  $\$u \setminus \text{GTSdAUR} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSdAUR}$   
 $\setminus \text{GTSnegatedDirectedAdjacencyUp}$  This macro will draw a negated directed adjacency upward symbol.  
 $\setminus \text{GTSnegDirAdjUp}$   $A. \setminus \text{GTSnegatedDirectedAdjacencyUp} / A.$  or  $A. \setminus \text{GTSnegDirAdjUp} / A.$  or  $A. \setminus \text{GTSnDAU} / A.$  will yield  $A. \setminus A.$   
 $\setminus \text{GTSnDAU}$   $u \setminus \text{GTSnegatedDirectedAdjacencyUp} / v$  or  $u \setminus \text{GTSnegDirAdjUp} / v$  or  $u \setminus \text{GTSnDAU} / v$  will yield  $u \setminus v$ .  
 $\$u \setminus \text{GTSnegatedDirectedAdjacencyUp} / v\$$  or  $\$u \setminus \text{GTSnegDirAdjUp} / v\$$  or  $\$u \setminus \text{GTSnDAU} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSnegatedDirectedAdjacencyUpRelation}$  This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.  
 $\setminus \text{GTSnegDirAdjUpRel}$   $\$u \setminus \text{GTSnegatedDirectedAdjacencyUpRelation} / v\$$  or  $\$u \setminus \text{GTSnegDirAdjUpRel} / v\$$  or  $\$u \setminus \text{GTSnDAUR} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSnDAUR}$   
 $\setminus \text{GTSdirectedAdjacencyDown}$  This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.  
 $\setminus \text{GTSdirAdjDown}$   $A. \setminus \text{GTSdirectedAdjacencyDown} / A.$  or  $A. \setminus \text{GTSdirAdjDown} / A.$  or  $A. \setminus \text{GTSnDAD} / A.$  will yield  $A. \setminus A.$   
 $\setminus \text{GTSdAD}$   $u \setminus \text{GTSdirectedAdjacencyDown} / v$  or  $u \setminus \text{GTSdirAdjDown} / v$  or  $u \setminus \text{GTSnDAD} / v$  will yield  $u \setminus v$ .  
 $\$u \setminus \text{GTSdirectedAdjacencyDown} / v\$$  or  $\$u \setminus \text{GTSdirAdjDown} / v\$$  or  $\$u \setminus \text{GTSnDAD} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSdirectedAdjacencyDownRelation}$  This macro will draw a directed adjacency downward symbol with relation spacing in math mode.  
 $\setminus \text{GTSdirAdjDownRel}$   $\$u \setminus \text{GTSdirectedAdjacencyDownRelation} / v\$$  or  $\$u \setminus \text{GTSdirAdjDownRel} / v\$$  or  $\$u \setminus \text{GTSdADR} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSdADR}$   
 $\setminus \text{GTSnegatedDirectedAdjacencyDown}$  This macro will draw a negated directed adjacency downward symbol.  
 $\setminus \text{GTSnegDirAdjDown}$   $A. \setminus \text{GTSnegatedDirectedAdjacencyDown} / A.$  or  $A. \setminus \text{GTSnegDirAdjDown} / A.$  or  $A. \setminus \text{GTSnDAD} / A.$  will yield  $A. \setminus A.$   
 $\setminus \text{GTSnDAD}$   $u \setminus \text{GTSnegatedDirectedAdjacencyDown} / v$  or  $u \setminus \text{GTSnegDirAdjDown} / v$  or  $u \setminus \text{GTSnDAD} / v$  will yield  $u \setminus v$ .  
 $\$u \setminus \text{GTSnegatedDirectedAdjacencyDown} / v\$$  or  $\$u \setminus \text{GTSnegDirAdjDown} / v\$$  or  $\$u \setminus \text{GTSnDAD} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSnegatedDirectedAdjacencyDownRelation}$  This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.  
 $\setminus \text{GTSnegDirAdjDownRel}$   $\$u \setminus \text{GTSnegatedDirectedAdjacencyDownRelation} / v\$$  or  $\$u \setminus \text{GTSnegDirAdjDownRel} / v\$$  or  $\$u \setminus \text{GTSnDADR} / v\$$  will yield  $u \setminus v$ .  
 $\setminus \text{GTSnDADR}$   
 $\setminus \text{GTS@i}$  This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and  $\setminus \text{GTS@j}$ :  $\setminus \text{GTSxincidence}$ ,  $\setminus \text{GTSxinc}$ , and  $\setminus \text{GTSxi}$ . The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.  
Arguments:  
 $\{ \langle mainColor \rangle \}$  If non-empty, this color will be used as main color.  
 $\{ \langle secondColor \rangle \}$  If non-empty, this color will be used as second color.  
 $\{ \langle drawSlash \rangle \}$  Non-empty implies that a negation will be drawn.  
 $\{ \langle drawUpArrow \rangle \}$  Non-empty implies that an up-arrow will be drawn.

`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

`\GTS@j` This macro builds on top of `\GTS@i` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@i`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.

`{\secondColor}` If non-empty, this color will be used as second color.

`{\drawSlash}` Non-empty implies that a negation will be drawn.

`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.

`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

`{\relation}` Non-empty implies that relation spacing will be added.

`\GTSxi` Versatile macros for incidence that can be given options.

`\GTSxinc` Argument:

`\GTSxincidence` [`\optionsList`] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,  
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,  
monochrome or mono or mon or m (equivalent to mC=black,sC=black),  
negated or not or n,  
in or i,  
out or o,  
relation or rel or r.

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

`\GTSinc` `A.\GTSincidence/A.` or `A.\GTSinc/A.` or `A.\GTSi/A.` will yield  $A.\downarrow A.$   
`v \GTSincidence/ e` or `v \GTSinc/ e` or `v \GTSi/ e` will yield  $v \downarrow e.$   
`$v \GTSincidence/ e$` or `$v \GTSinc/ e$` or `$v \GTSi/ e$` will yield  $v \downarrow e.$

`\GTSincidenceRelation` This macro will draw an incidence symbol with relation spacing in math mode.

`\GTSincRel` `$v\GTSincidenceRelation/e$` or `$v\GTSincRel/e$` or `$v\GTSiR/e$` will yield  $v \downarrow e.$

`\GTSiR`

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.

`\GTSnegInc` `A.\GTSnegatedIncidence/A.` or `A.\GTSnegInc/A.` or `A.\GTSnI/A.` will yield  $A.\not\downarrow A.$   
`v \GTSnegatedIncidence/ e` or `v \GTSnegInc/ e` or `v \GTSnI/ e` will yield  $v \not\downarrow e.$   
`$v \GTSnegatedIncidence/ e$` or `$v \GTSnegInc/ e$` or `$v \GTSnI/ e$` will yield  $v \not\downarrow e.$

`\GTSnegatedIncidence-` This macro will draw a negated incidence symbol with relation spacing in math mode.

Relation `$v\GTSnegatedIncidenceRelation/e$` or `$v\GTSnegIncRel/e$` or `$v\GTSnIR/e$` will yield  $v \not\downarrow e.$

`\GTSnegIncRel`

`\GTSnIR`

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`\GTSoutInc` `A.\GTSoutIncidence/A.` or `A.\GTSoutInc/A.` or `A.\GTSOI/A.` will yield  $A.\uparrow A.$   
`v \GTSoutIncidence/ e` or `v \GTSoutInc/ e` or `v \GTSOI/ e` will yield  $v \uparrow e.$   
`$v \GTSoutIncidence/ e$` or `$v \GTSoutInc/ e$` or `$v \GTSOI/ e$` will yield  $v \uparrow e.$

`\GTSoutIncidenceRelation` This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

`\GTSoutIncRel` `$v\GTSoutIncidenceRelation/e$` or `$v\GTSoutIncRel/e$` or `$v\GTSOIR/e$` will yield  $v \uparrow e.$

`\GTSOI`

`\GTSnegatedOutIncidence` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

`\GTSnegOutInc` `A.\GTSnegatedOutIncidence/A.` or `A.\GTSnegOutInc/A.` or `A.\GTSnOI/A.` will yield  $A.\not\uparrow A.$   
`v \GTSnegatedOutIncidence/ e` or `v \GTSnegOutInc/ e` or `v \GTSnOI/ e` will yield  $v \not\uparrow e.$   
`$v \GTSnegatedOutIncidence/ e$` or `$v \GTSnegOutInc/ e$` or `$v \GTSnOI/ e$` will yield  $v \not\uparrow e.$

`\GTSnegOutInc-` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

Relation `$v\GTSnegatedOutIncidenceRelation/e$` or `$v\GTSnegOutIncRel/e$` or `$v\GTSnOIR/e$` will yield  $v \not\uparrow e.$

`\GTSnOIR`

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`\GTSinInc`

`\GTSiI`

$A.\backslash\text{GTSinIncidence}/A.$  or  $A.\backslash\text{GTSinInc}/A.$  or  $A.\backslash\text{GTSiI}/A.$  will yield  $A.\text{\textcircled{A}}.$   
 $v \backslash\text{GTSinIncidence}/ e$  or  $v \backslash\text{GTSinInc}/ e$  or  $v \backslash\text{GTSiI}/ e$  will yield  $v \text{\textcircled{e}}.$   
 $\$v \backslash\text{GTSinIncidence}/ e\$$  or  $\$v \backslash\text{GTSinInc}/ e\$$  or  $\$v \backslash\text{GTSiI}/ e\$$  will yield  $v\text{\textcircled{e}}.$

$\backslash\text{GTSinIncidenceRelation}$  This macro will draw a directed incidence downward (in(ward) incidence) symbol with  
 $\backslash\text{GTSinIncRel}$  relation spacing in math mode.  
 $\backslash\text{GTSiIR}$   $\$v\backslash\text{GTSinIncidenceRelation}/e\$$  or  $\$v\backslash\text{GTSinIncRel}/e\$$  or  $\$v\backslash\text{GTSiIR}/e\$$  will yield  $v \text{\textcircled{e}}.$

$\backslash\text{GTSnegatedInIncidence}$  This macro will draw a negated directed incidence downward (in(ward) incidence) sym-  
 $\backslash\text{GTSnegInInc}$  bol.  
 $\backslash\text{GTSnII}$   $A.\backslash\text{GTSnegatedInIncidence}/A.$  or  $A.\backslash\text{GTSnegInInc}/A.$  or  $A.\backslash\text{GTSnII}/A.$  will yield  $A.\text{\textcircled{A}}.$   
 $v \backslash\text{GTSnegatedInIncidence}/ e$  or  $v \backslash\text{GTSnegInInc}/ e$  or  $v \backslash\text{GTSnII}/ e$  will yield  $v \text{\textcircled{e}}.$   
 $\$v \backslash\text{GTSnegatedInIncidence}/ e\$$  or  $\$v \backslash\text{GTSnegInInc}/ e\$$  or  $\$v \backslash\text{GTSnII}/ e\$$  will yield  $v\text{\textcircled{e}}.$

$\backslash\text{GTSnegatedInIncidence-}$  This macro will draw a negated directed incidence downward (in(ward) incidence)  
 $\text{Relation}$  symbol with relation spacing in math mode.  
 $\backslash\text{GTSnegInIncRel}$   $\$v\backslash\text{GTSnegatedInIncidenceRelation}/e\$$  or  $\$v\backslash\text{GTSnegInIncRel}/e\$$  or  $\$v\backslash\text{GTSnIIR}/e\$$   
 $\backslash\text{GTSnIIR}$  will yield  $v \text{\textcircled{e}}.$

$\backslash\text{GTSadjacencyMonochrome}$  This macro will draw a monochrome adjacency symbol.  
 $\backslash\text{GTSadjMono}$   $A.\backslash\text{GTSadjacencyMonochrome}/A.$  or  $A.\backslash\text{GTSadjMono}/A.$  or  $A.\backslash\text{GTSadjMon}/A.$  or  $A.\backslash\text{GTSaM}/A.$   
 $\backslash\text{GTSadjMon}$  will yield  $A.\text{\textcircled{A}}.$   
 $\backslash\text{GTSaM}$   $u \backslash\text{GTSadjacencyMonochrome}/ v$  or  $u \backslash\text{GTSadjMono}/ v$  or  $u \backslash\text{GTSadjMon}/ v$  or  $u \backslash\text{GTSaM}/$   
 $v$  will yield  $u \text{\textcircled{v}}.$   
 $\$u \backslash\text{GTSadjacencyMonochrome}/ v\$$  or  $\$u \backslash\text{GTSadjMono}/ v\$$  or  $\$u \backslash\text{GTSadjMon}/ v\$$  or  $\$u$   
 $\backslash\text{GTSaM}/ v\$$  will yield  $u\text{\textcircled{v}}.$

$\backslash\text{GTSadjacencyMonochrome-}$  This macro will draw a monochrome adjacency symbol with relation spacing in math  
 $\text{Relation}$  mode.  
 $\backslash\text{GTSadjMonoRel}$   $\$u\backslash\text{GTSadjacencyMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSadjMonoRel}/v\$$  or  $\$u\backslash\text{GTSadjMonRel}/v\$$   
 $\backslash\text{GTSadjMonRel}$  or  $\$u\backslash\text{GTSaMR}/v\$$  will yield  $u \text{\textcircled{v}}.$   
 $\backslash\text{GTSaMR}$

$\backslash\text{GTSunadjacencyMonochrome}$  This macro will draw a monochrome unadjacency symbol.  
 $\backslash\text{GTSunadjMono}$   $A.\backslash\text{GTSunadjacencyMonochrome}/A.$  or  $A.\backslash\text{GTSunadjMono}/A.$  or  $A.\backslash\text{GTSunadjMon}/A.$  or  
 $\backslash\text{GTSunadjMon}$   $A.\backslash\text{GTSuM}/A.$  will yield  $A.\text{\textcircled{A}}.$   
 $\backslash\text{GTSuM}$   $u \backslash\text{GTSunadjacencyMonochrome}/ v$  or  $u \backslash\text{GTSunadjMono}/ v$  or  $u \backslash\text{GTSunadjMon}/ v$  or  $u$   
 $\backslash\text{GTSuM}/ v$  will yield  $u \text{\textcircled{v}}.$   
 $\$u \backslash\text{GTSunadjacencyMonochrome}/ v\$$  or  $\$u \backslash\text{GTSunadjMono}/ v\$$  or  $\$u \backslash\text{GTSunadjMon}/$   
 $v\$$  or  $\$u \backslash\text{GTSuM}/ v\$$  will yield  $u\text{\textcircled{v}}.$

$\backslash\text{GTSunadjacencyMonochrome-}$  This macro will draw a monochrome unadjacency symbol with relation spacing in  
 $\text{Relation}$  math mode.  
 $\backslash\text{GTSunadjMonoRel}$   $\$u\backslash\text{GTSunadjacencyMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSunadjMonoRel}/v\$$  or  $\$u\backslash\text{GTSunadjMonRel}/v\$$   
 $\backslash\text{GTSunadjMonRel}$  or  $\$u\backslash\text{GTSuMR}/v\$$  will yield  $u \text{\textcircled{v}}.$   
 $\backslash\text{GTSuMR}$

$\backslash\text{GTSdirectedAdjacencyUp-}$  This macro will draw a monochrome directed adjacency upward symbol. The operand  
 $\text{Monochrome}$  on the left of the symbol should be considered the source of the arc; the operand on the  
 $\backslash\text{GTSdirAdjUpMono}$  right of the symbol should be considered the target of the arc.  
 $\backslash\text{GTSdirAdjUpMon}$   $A.\backslash\text{GTSdirectedAdjacencyUpMonochrome}/A.$  or  $A.\backslash\text{GTSdirAdjUpMono}/A.$  or  $A.\backslash\text{GTSdirAdjUpMon}/A.$   
 $\backslash\text{GTSdAUM}$  or  $A.\backslash\text{GTSdAUM}/A.$  will yield  $A.\text{\textcircled{A}}.$   
 $u \backslash\text{GTSdirectedAdjacencyUpMonochrome}/ v$  or  $u \backslash\text{GTSdirAdjUpMono}/ v$  or  $u \backslash\text{GTSdirAdjUpMon}/$   
 $v$  or  $u \backslash\text{GTSdAUM}/ v$  will yield  $u \text{\textcircled{v}}.$   
 $\$u \backslash\text{GTSdirectedAdjacencyUpMonochrome}/ v\$$  or  $\$u \backslash\text{GTSdirAdjUpMono}/ v\$$  or  $\$u \backslash\text{GTSdirAdjUpMon}/$   
 $v\$$  or  $\$u \backslash\text{GTSdAUM}/ v\$$  will yield  $u\text{\textcircled{v}}.$

$\backslash\text{GTSdirectedAdjacencyUp-}$  This macro will draw a monochrome directed adjacency upward symbol with relation  
 $\text{MonochromeRelation}$  spacing in math mode.  
 $\backslash\text{GTSdirAdjUpMonoRel}$   $\$u\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSdirAdjUpMonoRel}/v\$$  or  
 $\backslash\text{GTSdirAdjUpMonRel}$   $\$u\backslash\text{GTSdirAdjUpMonRel}/v\$$  or  $\$u\backslash\text{GTSdAUMR}/v\$$  will yield  $u \text{\textcircled{v}}.$   
 $\backslash\text{GTSdAUMR}$

$\backslash\text{GTSnegatedDirected-}$  This macro will draw a monochrome negated directed adjacency upward symbol.  
 $\text{AdjacencyUpMonochrome}$   $A.\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/A.$  or  $A.\backslash\text{GTSnegDirAdjUpMono}/A.$  or  
 $\backslash\text{GTSnegDirAdjUpMono}$   $A.\backslash\text{GTSnegDirAdjUpMon}/A.$  or  $A.\backslash\text{GTSnDAUM}/A.$  will yield  $A.\text{\textcircled{A}}.$   
 $\backslash\text{GTSnegDirAdjUpMon}$   $u \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/ v$  or  $u \backslash\text{GTSnegDirAdjUpMono}/ v$  or  $u$   
 $\backslash\text{GTSnDAUM}$   $\backslash\text{GTSnegDirAdjUpMon}/ v$  or  $u \backslash\text{GTSnDAUM}/ v$  will yield  $u \text{\textcircled{v}}.$   
 $\$u \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/ v\$$  or  $\$u \backslash\text{GTSnegDirAdjUpMono}/ v\$$

or  $\$u \backslash\text{GTSnegDirAdjUpMon}/ v\$$  or  $\$u \backslash\text{GTSnDAUM}/ v\$$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .

$\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}$  This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

$\text{Relation}$   $\$u \backslash\text{GTSnegatedDirectedAdjacencyUpMonochromeRelation}/ v\$$  or  $\$u \backslash\text{GTSnegDirAdjUpMonoRel}/ v\$$   
 $\backslash\text{GTSnegDirAdjUpMonoRel}$  or  $\$u \backslash\text{GTSnegDirAdjUpMonRel}/ v\$$  or  $\$u \backslash\text{GTSnDAUMR}/ v\$$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .  
 $\backslash\text{GTSnegDirAdjUpMonRel}$   
 $\backslash\text{GTSnDAUMR}$

$\backslash\text{GTSdirectedAdjacencyDownMonochrome}$  This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$\backslash\text{GTSdirAdjDownMono}$   $A \backslash\text{GTSdirectedAdjacencyDownMonochrome}/ A$ . or  $A \backslash\text{GTSdirAdjDownMono}/ A$ . or  $A \backslash\text{GTSdirAdjDownMono}$   
 $\backslash\text{GTSdADM}$  or  $A \backslash\text{GTSdADM}/ A$ . will yield  $A \overset{\circ}{\underset{\circ}{\parallel}} A$ .  
 $u \backslash\text{GTSdirectedAdjacencyDownMonochrome}/ v$  or  $u \backslash\text{GTSdirAdjDownMono}/ v$  or  $u \backslash\text{GTSdirAdjDownMono}$   
 $v$  or  $u \backslash\text{GTSdADM}/ v$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .

$\$u \backslash\text{GTSdirectedAdjacencyDownMonochrome}/ v\$$  or  $\$u \backslash\text{GTSdirAdjDownMono}/ v\$$  or  $\$u \backslash\text{GTSdirAdjDownMono}/ v\$$  or  $\$u \backslash\text{GTSdADM}/ v\$$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .

$\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}$  This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

$\backslash\text{GTSdirAdjDownMonoRel}$   $\$u \backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}/ v\$$  or  $\$u \backslash\text{GTSdirAdjDownMonoRel}/ v\$$   
 $\backslash\text{GTSdirAdjDownMonRel}$  or  $\$u \backslash\text{GTSdirAdjDownMonRel}/ v\$$  or  $\$u \backslash\text{GTSdADMR}/ v\$$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .  
 $\backslash\text{GTSdADMR}$

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}$  This macro will draw a monochrome negated directed adjacency downward symbol.  
 $A \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ A$ . or  $A \backslash\text{GTSnegDirAdjDownMono}/ A$ .  
 $\backslash\text{GTSnegDirAdjDownMono}$  or  $A \backslash\text{GTSnegDirAdjDownMon}/ A$ . or  $A \backslash\text{GTSnDADM}/ A$ . will yield  $A \overset{\circ}{\underset{\circ}{\parallel}} A$ .

$\backslash\text{GTSnegDirAdjDownMon}$   $u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ v$  or  $u \backslash\text{GTSnegDirAdjDownMono}/ v$   
 $\backslash\text{GTSnDADM}$  or  $u \backslash\text{GTSnegDirAdjDownMon}/ v$  or  $u \backslash\text{GTSnDADM}/ v$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .  
 $\$u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ v\$$  or  $\$u \backslash\text{GTSnegDirAdjDownMono}/ v\$$  or  $\$u \backslash\text{GTSnegDirAdjDownMon}/ v\$$  or  $\$u \backslash\text{GTSnDADM}/ v\$$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}$  This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

$\text{Relation}$   $\$u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}/ v\$$  or  $\$u \backslash\text{GTSnegDirAdjDownMonoRel}/ v\$$   
 $\backslash\text{GTSnegDirAdjDownMonoRel}$  or  $\$u \backslash\text{GTSnegDirAdjDownMonRel}/ v\$$  or  $\$u \backslash\text{GTSnDADMR}/ v\$$  will yield  $u \overset{\circ}{\underset{\circ}{\parallel}} v$ .  
 $\backslash\text{GTSnegDirAdjDownMonRel}$   
 $\backslash\text{GTSnDADMR}$

$\backslash\text{GTSincidenceMonochrome}$  This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

$\backslash\text{GTSincMono}$   $A \backslash\text{GTSincidenceMonochrome}/ A$ . or  $A \backslash\text{GTSincMono}/ A$ . or  $A \backslash\text{GTSincMon}/ A$ . or  $A \backslash\text{GTSiM}/ A$ .  
will yield  $A \downarrow A$ .

$v \backslash\text{GTSincidenceMonochrome}/ e$  or  $v \backslash\text{GTSincMono}/ e$  or  $v \backslash\text{GTSincMon}/ e$  or  $v \backslash\text{GTSiM}/ e$   
 $e$  will yield  $v \downarrow e$ .  
 $\$v \backslash\text{GTSincidenceMonochrome}/ e\$$  or  $\$v \backslash\text{GTSincMono}/ e\$$  or  $\$v \backslash\text{GTSincMon}/ e\$$  or  $\$v \backslash\text{GTSiM}/ e\$$  will yield  $v \downarrow e$ .

$\backslash\text{GTSincidenceMonochromeRelation}$  This macro will draw a monochrome incidence symbol with relation spacing in math mode.

$\text{Relation}$   $\$v \backslash\text{GTSincidenceMonochromeRelation}/ e\$$  or  $\$v \backslash\text{GTSincMonoRel}/ e\$$  or  $\$v \backslash\text{GTSincMonRel}/ e\$$   
 $\backslash\text{GTSincMonoRel}$  or  $\$v \backslash\text{GTSiMR}/ e\$$  will yield  $v \downarrow e$ .  
 $\backslash\text{GTSiMR}$

$\backslash\text{GTSnegatedIncidenceMonochrome}$  This macro will draw a monochrome negated incidence symbol.

$\text{Monochrome}$   $A \backslash\text{GTSnegatedIncidenceMonochrome}/ A$ . or  $A \backslash\text{GTSnegIncMono}/ A$ . or  $A \backslash\text{GTSnegIncMon}/ A$ .  
 $\backslash\text{GTSnegIncMono}$  or  $A \backslash\text{GTSnIM}/ A$ . will yield  $A \nmid A$ .

$\backslash\text{GTSnegIncMon}$   $v \backslash\text{GTSnegatedIncidenceMonochrome}/ e$  or  $v \backslash\text{GTSnegIncMono}/ e$  or  $v \backslash\text{GTSnegIncMon}/ e$   
 $\backslash\text{GTSnIM}$  or  $v \backslash\text{GTSnIM}/ e$  will yield  $v \nmid e$ .

$\$v \backslash\text{GTSnegatedIncidenceMonochrome}/ e\$$  or  $\$v \backslash\text{GTSnegIncMono}/ e\$$  or  $\$v \backslash\text{GTSnegIncMon}/ e\$$  or  $\$v \backslash\text{GTSnIM}/ e\$$  will yield  $v \nmid e$ .

$\backslash\text{GTSnegatedIncidenceMonochromeRelation}$  This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

$\text{Relation}$   $\$v \backslash\text{GTSnegatedIncidenceMonochromeRelation}/ e\$$  or  $\$v \backslash\text{GTSnegIncMonoRel}/ e\$$  or  $\$v \backslash\text{GTSnegIncMonRel}/ e\$$   
 $\backslash\text{GTSnegIncMonoRel}$  or  $\$v \backslash\text{GTSnIMR}/ e\$$  will yield  $v \nmid e$ .  
 $\backslash\text{GTSnIMR}$



`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`\GTSoutIncMono` `A.\GTSoutIncidenceMonochrome/A.` or `A.\GTSoutIncMono/A.` or `A.\GTSoutIncMon/A.` or `A.\GTSoutIM/A.` will yield  $A.\textcircled{\uparrow}A.$

`v \GTSoutIncidenceMonochrome/ e` or `v \GTSoutIncMono/ e` or `v \GTSoutIncMon/ e` or `v \GTSoutIM/ e` will yield  $v \textcircled{\uparrow} e.$

`$v \GTSoutIncidenceMonochrome/ e$` or `$v \GTSoutIncMono/ e$` or `$v \GTSoutIncMon/ e$` or `$v \GTSoutIM/ e$` will yield  $v \textcircled{\uparrow} e.$

`\GTSoutIncidenceMonochrome-Relation` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

`\GTSoutIncMonoRel` `$v \GTSoutIncidenceMonochromeRelation/e$` or `$v \GTSoutIncMonoRel/e$` or `$v \GTSoutIncMonRel/e$` or `$v \GTSoutIMR/e$` will yield  $v \textcircled{\uparrow} e.$

`\GTSoutIncMonRel`

`\GTSoutIMR`

`\GTSnegatedOutIncidenceMonochrome` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

`\GTSnegOutIncMono` `A.\GTSnegatedOutIncidenceMonochrome/A.` or `A.\GTSnegOutIncMono/A.` or `A.\GTSnegOutIncMon/A.` or `A.\GTSnOIM/A.` will yield  $A.\textcircled{\uparrow}A.$

`\GTSnegOutIncMon` `v \GTSnegatedOutIncidenceMonochrome/ e` or `v \GTSnegOutIncMono/ e` or `v \GTSnegOutIncMon/ e` or `v \GTSnOIM/ e` will yield  $v \textcircled{\uparrow} e.$

`$v \GTSnegatedOutIncidenceMonochrome/ e$` or `$v \GTSnegOutIncMono/ e$` or `$v \GTSnegOutIncMon/ e$` or `$v \GTSnOIM/ e$` will yield  $v \textcircled{\uparrow} e.$

`\GTSnegatedOutIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

`\GTSnegOutIncMonoRel` `$v \GTSnegatedOutIncidenceMonochromeRelation/e$` or `$v \GTSnegOutIncMonoRel/e$` or `$v \GTSnegOutIncMonRel/e$` or `$v \GTSnOIMR/e$` will yield  $v \textcircled{\uparrow} e.$

`\GTSnegOutIncMonRel`

`\GTSnOIMR`

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`\GTSinIncMono` `A.\GTSinIncidenceMonochrome/A.` or `A.\GTSinIncMono/A.` or `A.\GTSinIncMon/A.` or `A.\GTSiIM/A.` will yield  $A.\textcircled{\downarrow}A.$

`v \GTSinIncidenceMonochrome/ e` or `v \GTSinIncMono/ e` or `v \GTSinIncMon/ e` or `v \GTSiIM/ e` will yield  $v \textcircled{\downarrow} e.$

`$v \GTSinIncidenceMonochrome/ e$` or `$v \GTSinIncMono/ e$` or `$v \GTSinIncMon/ e$` or `$v \GTSiIM/ e$` will yield  $v \textcircled{\downarrow} e.$

`\GTSinIncidenceMonochrome-Relation` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSinIncMonoRel` `$v \GTSinIncidenceMonochromeRelation/e$` or `$v \GTSinIncMonoRel/e$` or `$v \GTSinIncMonRel/e$` or `$v \GTSiIMR/e$` will yield  $v \textcircled{\downarrow} e.$

`\GTSinIncMonRel`

`\GTSiIMR`

`\GTSnegatedInIncidenceMonochrome` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

`\GTSnegInIncMono` `A.\GTSnegatedInIncidenceMonochrome/A.` or `A.\GTSnegInIncMono/A.` or `A.\GTSnegInIncMon/A.` or `A.\GTSnIIM/A.` will yield  $A.\textcircled{\downarrow}A.$

`\GTSnegInIncMon` `v \GTSnegatedInIncidenceMonochrome/ e` or `v \GTSnegInIncMono/ e` or `v \GTSnegInIncMon/ e` or `v \GTSnIIM/ e` will yield  $v \textcircled{\downarrow} e.$

`$v \GTSnegatedInIncidenceMonochrome/ e$` or `$v \GTSnegInIncMono/ e$` or `$v \GTSnegInIncMon/ e$` or `$v \GTSnIIM/ e$` will yield  $v \textcircled{\downarrow} e.$

`\GTSnegatedInIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSnegInIncMonoRel` `$v \GTSnegatedInIncidenceMonochromeRelation/e$` or `$v \GTSnegInIncMonoRel/e$` or `$v \GTSnegInIncMonRel/e$` or `$v \GTSnIIMR/e$` will yield  $v \textcircled{\downarrow} e.$

`\GTSnegInIncMonRel`

`\GTSnIIMR`

### 3 Implementation

```

1 <*package>
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}

```

```

5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}
14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17

```

**\GTS@newcommand** This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

$\langle commandname \rangle$  The name of the command to create.

$\langle commandcode \rangle$  The code of the command to create.

```

18 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
19   \@ifdefinable{#1}{\def#1/{#2}}
20 }}

```

**\GTSsetMainColor** This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

$\langle color \rangle$  The color that must be used by GraphTheorySymbol afterward.

```

21 \newcommand{\GTSsetMainColor}[1]{%
22 \edef\GTS@mainColorTemp{#1}%
23 \edef\GTS@mainColorBackup{\GTS@mainColor}%
24 \edef\GTS@mainColor{\GTS@mainColorTemp}%
25 }

```

**\GTSsetSecondColor** This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

$\langle color \rangle$  The color that must be used by GraphTheorySymbol afterward.

```

26 \newcommand{\GTSsetSecondColor}[1]{%
27 \edef\GTS@secondColorTemp{#1}%
28 \edef\GTS@secondColorBackup{\GTS@secondColor}%
29 \edef\GTS@secondColor{\GTS@secondColorTemp}%
30 }

```

**\GTS@a** This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical. See the macros derived from it and **\GTS@b**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

$\langle mainColor \rangle$  If non-empty, this color will be used as main color.

$\langle secondColor \rangle$  If non-empty, this color will be used as second color.

$\langle drawSlash \rangle$  Non-empty implies that a negation will be drawn.

$\langle drawUpArrow \rangle$  Non-empty implies that an up-arrow will be drawn.

$\langle drawDownArrow \rangle$  Non-empty implies that a down-arrow will be drawn.

```

31 \newcommand{\GTS@a}[5]{%
32 \if#1\empty\else%
33 \GTSsetMainColor{#1}%
34 \fi%
35 \if#2\empty\else%
36 \GTSsetSecondColor{#2}%
37 \fi%
38 \text{\ensuremath{%
39 \raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
40 %
41 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
42 $\mathcolor{\GTS@secondColor}{\circ}$%

```

```

43 }}%
44 \hspace{-0.39ex}%
45 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
46 $\mathcolor{\GTS@secondColor}{\bullet}$%
47 }}%
48 \hspace{-0.12ex}%
49 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
50 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
51 }}%
52 \hspace{-0.40ex}%
53 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
54 $\mathcolor{\GTS@secondColor}{\circ}$%
55 }}%
56 \hspace{-0.39ex}%
57 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
58 $\mathcolor{\GTS@secondColor}{\bullet}$%
59 }}%
60 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
61 \if#4\empty\else%
62 \hspace{-0.65ex}%
63 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
64 \textcolor{\GTS@mainColor}{\textasciicircum}%
65 }}%
66 \fi%
if GTS@drawDownArrow empty else
67 \if#5\empty\else%
68 \hspace{-0.65ex}%
69 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
70 \textcolor{\GTS@mainColor}{\textasciicircum}%
71 }}%
72 \fi%
if GTS@drawSlash empty else
73 \if#3\empty\else%
74 \hspace{-0.65ex}%
75 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
76 \textcolor{black}{/}%
77 }}%
78 \fi%
79 %
80 }}%
81 }}%
82 \if#1\empty\else%
83 \GTSsetMainColor{\GTS@mainColorBackup}%
84 \fi%
85 \if#2\empty\else%
86 \GTSsetSecondColor{\GTS@secondColorBackup}%
87 \fi%
88 }

```

**\GTS@b** This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

- {<mainColor>}** If non-empty, this color will be used as main color.
- {<secondColor>}** If non-empty, this color will be used as second color.
- {<drawSlash>}** Non-empty implies that a negation will be drawn.
- {<drawUpArrow>}** Non-empty implies that an up-arrow will be drawn.
- {<drawDownArrow>}** Non-empty implies that a down-arrow will be drawn.
- {<relation>}** Non-empty implies that relation spacing will be added.

```

89 \newcommand{\GTS@b}[6]{%
90 \if#61%
91 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
92 \else%

```

```

93 \GTS@a{#1}{#2}{#3}{#4}{#5}%
94 \fi%
95 }

```

`\GTSxa` Versatile macros for adjacency that can be given options.

`\GTSxadj` Argument:

`\GTSxadjacency` [*optionsList*] Sets the following options if not empty:  
 mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,  
 secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,  
 monochrome or mono or mon or m (equivalent to mC=black,sC=black),  
 negated or not or n,  
 up or u,  
 down or d,  
 relation or rel or r.

```

96 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
97 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
98 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
99 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
101 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
102 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
103 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{monochrome}[1]%
105 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
106 \define@key{GTSxa}{mono}[1]%
107 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
108 \define@key{GTSxa}{mon}[1]%
109 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
110 \define@key{GTSxa}{m}[1]%
111 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
112 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
113 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
114 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
115 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
116 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
117 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
118 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
119 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
120 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
121 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
122
123 \newcommand{\GTSxa}[1] [] {%
124 \def\GTS@@mainColor{}%
125 \def\GTS@@secondColor{}%
126 \def\GTS@@drawSlash{}%
127 \def\GTS@@drawUpArrow{}%
128 \def\GTS@@drawDownArrow{}%
129 \def\GTS@@relation{}%
130 \setkeys{GTSxa}{#1}%
131 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
132 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
133 }
134
135 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
136
137 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

`\GTSadjacency` This macro will draw an adjacency symbol.

`\GTSadj` 138 `\GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}{}`

`\GTSa` 139 `\GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}{}`

140 `\GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}{}`

`\GTSadjacencyRelation` This macro will draw an adjacency symbol with relation spacing in math mode.

`\GTSadjRel` 141 `\GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}`

`\GTSaR` 142 `\GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}`

143 `\GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}`

`\GTSunadjacency` This macro will draw an unadjacency symbol.

```

\GTSunadj 144 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}}
\GTSu     145 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}}
          146 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}}

```

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol with relation spacing in math mode.

```

\GTSunadjRel 147 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
\GTSuR       148 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
          149 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

          150 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}}
          151 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}}
          152 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}}

```

`\GTSdirectedAdjacencyUpRelation` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

\GTSdirAdjUpRel 153 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
\GTSdAUR       154 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
          155 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

`\GTSnegatedDirectedAdjacencyUp` This macro will draw a negated directed adjacency upward symbol.

```

          156 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}}
\GTSnegDirAdjUp 157 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}}
\GTSnDAU       158 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}}

```

`\GTSnegatedDirectedAdjacencyUpRelation` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

\GTSnegDirAdjUpRel 159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
\GTSnDAUR       160 \mathrel{\GTSnDAU/}%
          161 }
          162 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
          163 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

```

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

          164 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}}
          165 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}}
          166 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}}

```

`\GTSdirectedAdjacencyDownRelation` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdirAdjDownRel 167 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
\GTSdADR       168 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
          169 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

```

`\GTSnegatedDirectedAdjacencyDown` This macro will draw a negated directed adjacency downward symbol.

```

          170 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{1}}
\GTSnegDirAdjDown 171 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{1}}
\GTSnDAD       172 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{1}}

```

`\GTSnegatedDirectedAdjacencyDownRelation` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

\GTSnegDirAdjDownRel 173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
\GTSnDADR       174 \mathrel{\GTSnDAD/}%
          175 }
          176 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
          177 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

```

`\GTS@i` This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.

`{\secondColor}` If non-empty, this color will be used as second color.

`{\drawSlash}` Non-empty implies that a negation will be drawn.

`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.

`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

```

178 \newcommand{\GTS@i}[5]{%
179 \if#1\empty\else%
180 \GTSsetMainColor{#1}%
181 \fi%
182 \if#2\empty\else%
183 \GTSsetSecondColor{#2}%
184 \fi%
185 \text{\ensuremath{%
186 \raisebox{0ex}{\scalebox{1.8}[1.8]{%
187 %
188 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
189 $\mathcolor{\GTS@secondColor}{\circ}$%
190 }}%
191 \hspace{-0.39ex}%
192 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
193 $\mathcolor{\GTS@secondColor}{\bullet}$%
194 }}%
195 \hspace{-0.12ex}%
196 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
197 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
198 }}%
199 \hspace{-0.18ex}%
200 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
201 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
202 }}%
203 \hspace{0.2ex}%
if GTS@drawUpArrow empty else
204 \if#4\empty\else%
205 \hspace{-0.65ex}%
206 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
207 \textcolor{\GTS@secondColor}{\textasciicircum}%
208 }}%
209 \fi%
if GTS@drawDownArrow empty else
210 \if#5\empty\else%
211 \hspace{-0.65ex}%
212 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
213 \textcolor{\GTS@secondColor}{\textasciicircum}%
214 }}%
215 \fi%
if GTS@drawSlash empty else
216 \if#3\empty\else%
217 \hspace{-0.65ex}%
218 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
219 \textcolor{black}{/}%
220 }}%
221 \fi%
222 %
223 }}%
224 }}%
225 \if#1\empty\else%
226 \GTSsetMainColor{\GTS@mainColorBackup}%
227 \fi%
```

```

228 \if#2\empty\else%
229 \GTSsetSecondColor{\GTS@secondColorBackup}%
230 \fi%
231 }

```

**\GTS@j** This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

**{\mainColor}** If non-empty, this color will be used as main color.  
**{\secondColor}** If non-empty, this color will be used as second color.  
**{\drawSlash}** Non-empty implies that a negation will be drawn.  
**{\drawUpArrow}** Non-empty implies that an up-arrow will be drawn.  
**{\drawDownArrow}** Non-empty implies that a down-arrow will be drawn.  
**{\relation}** Non-empty implies that relation spacing will be added.

```

232 \newcommand{\GTS@j}[6]{%
233 \if#61%
234 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
235 \else%
236 \GTS@i{#1}{#2}{#3}{#4}{#5}%
237 \fi%
238 }

```

**\GTSxi** Versatile macros for incidence that can be given options.

**\GTSxinc** Argument:

**\GTSxincidence** [*optionsList*] Sets the following options if not empty:  
 mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,  
 secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,  
 monochrome or mono or mon or m (equivalent to mC=black,sC=black),  
 negated or not or n,  
 in or i,  
 out or o,  
 relation or rel or r.

```

239 \define@key{GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
240 \define@key{GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
241 \define@key{GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
242 \define@key{GTSxi}{mC}{\def\GTS@@mainColor{#1}}
243 \define@key{GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
244 \define@key{GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
245 \define@key{GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
246 \define@key{GTSxi}{sC}{\def\GTS@@secondColor{#1}}
247 \define@key{GTSxi}{monochrome}[1]%
248 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
249 \define@key{GTSxi}{mono}[1]%
250 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
251 \define@key{GTSxi}{mon}[1]%
252 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
253 \define@key{GTSxi}{m}[1]%
254 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
255 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
256 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
257 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
258 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
259 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
260 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
261 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
262 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
263 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
264 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
265
266 \newcommand{\GTSxi}[1][]{%
267 \def\GTS@@mainColor{}%
268 \def\GTS@@secondColor{}%
269 \def\GTS@@drawSlash{}%

```

```

270 \def\GTS@@drawUpArrow{%
271 \def\GTS@@drawDownArrow{%
272 \def\GTS@@relation{%
273 \setkeys{GTSxi}{#1}%
274 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
275 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
276 }
277
278 \newcommand{\GTSxinc}[1][\GTSxi[#1]]{
279
280 \newcommand{\GTSxincidence}[1][\GTSxi[#1]]{

\GTSincidence This macro will draw an incidence symbol. The operand on the left of the symbol should be
\GTSinc considered the vertex to which the edge/the operand on the right of the symbol is incident.
\GTSi 281 \GTS@newcommand{\GTSincidence}{\GTS@i{}{}{}{}}
282 \GTS@newcommand{\GTSinc}{\GTS@i{}{}{}{}}
283 \GTS@newcommand{\GTSi}{\GTS@i{}{}{}{}}

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.
\GTSincRel 284 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
\GTSiR 285 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
286 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

\GTSnegatedIncidence This macro will draw a negated incidence symbol.
\GTSnegInc 287 \GTS@newcommand{\GTSnegatedIncidence}{\GTS@i{}{}{1}{}{}}
\GTSnI 288 \GTS@newcommand{\GTSnegInc}{\GTS@i{}{}{1}{}{}}
289 \GTS@newcommand{\GTSnI}{\GTS@i{}{}{1}{}{}}

\GTSnegatedIncidence- This macro will draw a negated incidence symbol with relation spacing in math mode.
Relation 290 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
\GTSnegIncRel 291 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
\GTSnIR 292 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

\GTSoutIncidence This macro will draw a directed incidence upward (out(ward) incidence) symbol. The
\GTSoutInc operand on the left of the symbol should be considered the vertex to which the arc/the
\GTSOI operand on the right of the symbol is incident.
293 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{}{1}{}{}}
294 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{}{1}{}{}}
295 \GTS@newcommand{\GTSOI}{\GTS@i{}{}{}{1}{}{}}

\GTSoutIncidenceRelation This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation
\GTSoutIncRel spacing in math mode.
\GTSOIR 296 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSOI/}}
297 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSOI/}}
298 \GTS@newcommand{\GTSOIR}{\mathrel{\GTSOI/}}

\GTSnegatedOutIncidence This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.
\GTSnegOutInc 299 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}{}}
\GTSnOI 300 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}{}}
301 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}{}}

\GTSnegatedOutIncidence- This macro will draw a negated directed incidence upward (out(ward) incidence) symbol
Relation with relation spacing in math mode.
\GTSnegOutIncRel 302 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
\GTSnOIR 303 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
304 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}

\GTSinIncidence This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
\GTSinInc operand on the left of the symbol should be considered the vertex to which the arc/the
\GTSiI operand on the right of the symbol is incident.
305 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{}{1}}
306 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{}{1}}
307 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{}{1}}

```



`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSinIncRel`

```

\GTSiIR 308 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
309 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
310 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

`\GTSnegInInc`

```

\GTSnII 311 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{}{1}}
312 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{}{1}}
313 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{}{1}}

```

`\GTSnegatedInIncidence-Relation` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSnegInIncRel`

```

\GTSnIIR 314 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
315 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
316 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}

```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

`\GTSadjMono`

```

\GTSadjMon 317 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}{}{}}
318 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}{}{}}
\GTSaM 319 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}{}{}}
320 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}{}{}}

```

`\GTSadjacencyMonochrome-Relation` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

`\GTSadjMonoRel`

```

\GTSadjMonRel 321 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
322 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
323 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
\GTSaMR 324 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}

```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

`\GTSunadjMono`

```

\GTSunadjMon 325 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}{}}
326 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}{}}
\GTSuM 327 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}{}}
328 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}{}}

```

`\GTSunadjacencyMonochrome-Relation` This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

`\GTSunadjMonoRel`

```

\GTSunadjMonRel 329 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
330 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
\GTSuMR 331 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
332 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}

```

`\GTSdirectedAdjacencyUp-Monochrome` This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

`\GTSdirAdjUpMono`

```

\GTSdirAdjUpMon 333 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
\GTSdAUM 334 \GTS@a{black}{black}{}{1}{}%
335 }
336 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{}{1}{}{}}
337 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{}{1}{}{}}
338 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{}{1}{}{}}

```

`\GTSdirectedAdjacencyUp-MonochromeRelation` This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

`\GTSdirAdjUpMonoRel`

```

\GTSdirAdjUpMonRel 339 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
\GTSdAUMR 340 \mathrel{\GTSdAUM/}%
341 }
342 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
343 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
344 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}

```

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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| $\backslash$ GTSnegatedDirected-<br>AdjacencyUpMonochrome-<br>$\backslash$ GTSnegDirAdjUpMono<br>$\backslash$ GTSnegDirAdjUpMon<br>$\backslash$ GTSnDAUM<br><br>$\backslash$ GTSnegatedDirected-<br>AdjacencyUpMonochrome-<br>Relation<br>$\backslash$ GTSnegDirAdjUpMonRel<br>$\backslash$ GTSnegDirAdjUpMonoRel<br>$\backslash$ GTSnDAUMR<br><br>$\backslash$ GTSdirectedAdjacencyDown-<br>Monochrome<br>$\backslash$ GTSdirAdjDownMono<br>$\backslash$ GTSdirAdjDownMon<br>$\backslash$ GTSdADM<br><br>$\backslash$ GTSdirectedAdjacencyDown-<br>MonochromeRelation<br>$\backslash$ GTSdirAdjDownMonoRel<br>$\backslash$ GTSdirAdjDownMonRel<br>$\backslash$ GTSdADMR<br><br>$\backslash$ GTSnegatedDirected-<br>AdjacencyDownMonochrome<br>$\backslash$ GTSnegDirAdjDownMono<br>$\backslash$ GTSnegDirAdjDownMon<br>$\backslash$ GTSnDADM<br><br>$\backslash$ GTSnegatedDirected-<br>AdjacencyDownMonochrome-<br>Relation<br>$\backslash$ GTSnegDirAdjDownMonoRel<br>$\backslash$ GTSnegDirAdjDownMonRel<br>$\backslash$ GTSnDADMR<br><br>$\backslash$ GTSincidenceMonochrome<br>$\backslash$ GTSincMono<br>$\backslash$ GTSincMon<br>$\backslash$ GTSiM<br><br>$\backslash$ GTSincidenceMonochrome-<br>Relation<br>$\backslash$ GTSincMonoRel<br>$\backslash$ GTSincMonRel<br>$\backslash$ GTSiMR | <p>This macro will draw a monochrome negated directed adjacency upward symbol.</p> <p>345 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegatedDirectedAdjacencyUpMonochrome<math>\}</math>{%</p> <p>346 <math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p> <p>347 }</p> <p>348 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjUpMono<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p> <p>349 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjUpMon<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p> <p>350 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnDAUM<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p><br><p>This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.</p> <p>351 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegatedDirectedAdjacencyUpMonochromeRelation<math>\}</math>{%</p> <p>352 <math>\mathrel{\backslash</math>GTSnDAUM/}</p> <p>353 }</p> <p>354 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjUpMonoRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSnDAUM/}}</p> <p>355 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjUpMonRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSnDAUM/}}</p> <p>356 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnDAUMR<math>\}</math>{<math>\mathrel{\backslash</math>GTSnDAUM/}}</p><br><p>This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.</p> <p>357 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdirectedAdjacencyDownMonochrome<math>\}</math>{%</p> <p>358 <math>\backslash</math>GTS@a{black}{black}{}</p> <p>359 }</p> <p>360 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdirAdjDownMono<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{}</p> <p>361 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdirAdjDownMon<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{}</p> <p>362 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdADM<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{}</p><br><p>This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.</p> <p>363 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdirectedAdjacencyDownMonochromeRelation<math>\}</math>{%</p> <p>364 <math>\mathrel{\backslash</math>GTSdADM/}</p> <p>365 }</p> <p>366 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdirAdjDownMonoRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSdADM/}}</p> <p>367 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdirAdjDownMonRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSdADM/}}</p> <p>368 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSdADMR<math>\}</math>{<math>\mathrel{\backslash</math>GTSdADM/}}</p><br><p>This macro will draw a monochrome negated directed adjacency downward symbol.</p> <p>369 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegatedDirectedAdjacencyDownMonochrome<math>\}</math>{%</p> <p>370 <math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p> <p>371 }</p> <p>372 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjDownMono<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p> <p>373 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjDownMon<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p> <p>374 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnDADM<math>\}</math>{<math>\backslash</math>GTS@a{black}{black}{1}{1}{}}</p><br><p>This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.</p> <p>375 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegatedDirectedAdjacencyDownMonochromeRelation<math>\}</math>{%</p> <p>376 <math>\mathrel{\backslash</math>GTSnDADM/}</p> <p>377 }</p> <p>378 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjDownMonoRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSnDADM/}}</p> <p>379 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnegDirAdjDownMonRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSnDADM/}}</p> <p>380 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSnDADMR<math>\}</math>{<math>\mathrel{\backslash</math>GTSnDADM/}}</p><br><p>This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.</p> <p>381 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSincidenceMonochrome<math>\}</math>{<math>\backslash</math>GTS@i{black}{black}{}</p> <p>382 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSincMono<math>\}</math>{<math>\backslash</math>GTS@i{black}{black}{}</p> <p>383 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSincMon<math>\}</math>{<math>\backslash</math>GTS@i{black}{black}{}</p> <p>384 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSiM<math>\}</math>{<math>\backslash</math>GTS@i{black}{black}{}</p><br><p>This macro will draw a monochrome incidence symbol with relation spacing in math mode.</p> <p>385 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSincidenceMonochromeRelation<math>\}</math>{<math>\mathrel{\backslash</math>GTSiM/}}</p> <p>386 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSincMonoRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSiM/}}</p> <p>387 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSincMonRel<math>\}</math>{<math>\mathrel{\backslash</math>GTSiM/}}</p> <p>388 <math>\backslash</math>GTS@newcommand<math>\{\backslash</math>GTSiMR<math>\}</math>{<math>\mathrel{\backslash</math>GTSiM/}}</p> |
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`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

Monochrome 389 `\GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%`  
`\GTSnegIncMono` 390 `\GTS@i{black}{black}{1}{-}{-}%`  
`\GTSnegIncMon` 391 `}`  
`\GTSnIM` 392 `\GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{-}{-}}`  
393 `\GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{-}{-}}`  
394 `\GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{-}{-}}`

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.  
MonochromeRelation  
`\GTSnegIncMonoRel` 395 `\GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}`  
`\GTSnegIncMonRel` 396 `\GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}`  
`\GTSnIMR` 397 `\GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}`  
398 `\GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}`

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the  
`\GTSoutIncMono` arc/the operand on the right of the symbol is incident.  
`\GTSoutIncMon`  
`\GTSoIM` 399 `\GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{-}{1}{-}}`  
400 `\GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{-}{1}{-}}`  
401 `\GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{-}{1}{-}}`  
402 `\GTS@newcommand{\GTSoIM}{\GTS@i{black}{black}{-}{1}{-}}`

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.  
Relation  
`\GTSoutIncMonoRel` 403 `\GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoIM/}}`  
`\GTSoutIncMonRel` 404 `\GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoIM/}}`  
`\GTSoIMR` 405 `\GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoIM/}}`  
406 `\GTS@newcommand{\GTSoIMR}{\mathrel{\GTSoIM/}}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.  
Monochrome  
`\GTSnegOutIncMono` 407 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%`  
`\GTSnegOutIncMon` 408 `\GTS@i{black}{black}{1}{1}{-}%`  
`\GTSnOIM` 409 `}`  
410 `\GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}{-}}`  
411 `\GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}{-}}`  
412 `\GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}{-}}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.  
MonochromeRelation  
`\GTSnegOutIncMonoRel` 413 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%`  
`\GTSnegOutIncMonRel` 414 `\mathrel{\GTSnOIM/}%`  
`\GTSnOIMR` 415 `}`  
416 `\GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}`  
417 `\GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}`  
418 `\GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}`

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the  
`\GTSinIncMono` arc/the operand on the right of the symbol is incident.  
`\GTSinIncMon`  
`\GTSiIM` 419 `\GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{-}{-}{1}}`  
420 `\GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{-}{-}{1}}`  
421 `\GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{-}{-}{1}}`  
422 `\GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{-}{-}{1}}`

`\GTSinIncidenceMonochrome-` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.  
Relation  
`\GTSinIncMonoRel` 423 `\GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}`  
`\GTSinIncMonRel` 424 `\GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}`  
`\GTSiIMR` 425 `\GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}`  
426 `\GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}`

```

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
Monochrome dence) symbol.
\GTSnegInIncMono 427 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
\GTSnegInIncMon 428 \GTS@i{black}{black}{1}{1}%
\GTSnIIM 429 }
430 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{1}}
431 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{1}}
432 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{1}}

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
MonochromeRelation dence) symbol with relation spacing in math mode.
\GTSnegInIncMonoRel 433 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
\GTSnegInIncMonRel 434 \mathrel{\GTSnIIM/}%
\GTSnIIMR 435 }
436 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
437 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
438 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}

439 \endinput
440 </package>

```

## 4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

## 5 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

## 6 Change History

|                                                            |                                                                      |
|------------------------------------------------------------|----------------------------------------------------------------------|
| v0.1.0 – 2026-08-01                                        | v1.2.0 – 2026-09-06                                                  |
| General: Initial version . . . . . 1                       | General: CTAN URL, new macros for<br>same set of symbols . . . . . 1 |
| v1.0.0 – 2026-08-26                                        |                                                                      |
| General: First public release . . . . . 1                  | v1.2.1 – 2026-09-08                                                  |
| v1.1.0 – 2026-08-29                                        | General: smaller tests output . . . . . 1                            |
| General: Added a memo . . . . . 1                          | v1.3.0 – 2026-09-08                                                  |
| v1.1.1 – 2026-08-30                                        | General: new macros for monochrome<br>symbols . . . . . 1            |
| General: Small corrections and<br>improvements . . . . . 1 |                                                                      |

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| \GTS@b .....                              | 4, <a href="#">89</a> , <a href="#">131</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| \GTS@drawDownArrow .....                  | <a href="#">16</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| \GTS@drawSlash .....                      | <a href="#">14</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| \GTS@drawUpArrow .....                    | <a href="#">15</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| \GTS@i ..                                 | 5, <a href="#">234</a> , <a href="#">236</a> , <a href="#">281</a> , <a href="#">282</a> , <a href="#">283</a> , <a href="#">287</a> , <a href="#">288</a> , <a href="#">289</a> , <a href="#">293</a> , <a href="#">294</a> , <a href="#">295</a> , <a href="#">299</a> , <a href="#">300</a> , <a href="#">301</a> , <a href="#">305</a> , <a href="#">306</a> ,<br><a href="#">307</a> , <a href="#">311</a> , <a href="#">312</a> , <a href="#">313</a> , <a href="#">381</a> , <a href="#">382</a> , <a href="#">383</a> , <a href="#">384</a> , <a href="#">390</a> , <a href="#">392</a> , <a href="#">393</a> , <a href="#">394</a> , <a href="#">399</a> , <a href="#">400</a> , <a href="#">401</a> , <a href="#">402</a> , <a href="#">408</a> ,<br><a href="#">410</a> , <a href="#">411</a> , <a href="#">412</a> , <a href="#">419</a> , <a href="#">420</a> , <a href="#">421</a> , <a href="#">422</a> , <a href="#">428</a> , <a href="#">430</a> , <a href="#">431</a> , <a href="#">432</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| \GTS@j .....                              | 6, <a href="#">274</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTS@mainColor .....                      | 10, <a href="#">23</a> , <a href="#">24</a> , <a href="#">50</a> , <a href="#">64</a> , <a href="#">70</a> , <a href="#">201</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| \GTS@mainColorBackup .....                | 12, <a href="#">23</a> , <a href="#">83</a> , <a href="#">226</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| \GTS@mainColorTemp .....                  | 22, <a href="#">24</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTS@newcommand ..                        | 4, <a href="#">18</a> , <a href="#">138</a> , <a href="#">139</a> , <a href="#">140</a> , <a href="#">141</a> , <a href="#">142</a> , <a href="#">143</a> , <a href="#">144</a> , <a href="#">145</a> , <a href="#">146</a> , <a href="#">147</a> , <a href="#">148</a> , <a href="#">149</a> , <a href="#">150</a> ,<br><a href="#">151</a> , <a href="#">152</a> , <a href="#">153</a> , <a href="#">154</a> , <a href="#">155</a> , <a href="#">156</a> , <a href="#">157</a> , <a href="#">158</a> , <a href="#">159</a> , <a href="#">162</a> , <a href="#">163</a> , <a href="#">164</a> , <a href="#">165</a> , <a href="#">166</a> , <a href="#">167</a> , <a href="#">168</a> , <a href="#">169</a> ,<br><a href="#">170</a> , <a href="#">171</a> , <a href="#">172</a> , <a href="#">173</a> , <a href="#">176</a> , <a href="#">177</a> , <a href="#">281</a> , <a href="#">282</a> , <a href="#">283</a> , <a href="#">284</a> , <a href="#">285</a> , <a href="#">286</a> , <a href="#">287</a> , <a href="#">288</a> , <a href="#">289</a> , <a href="#">290</a> , <a href="#">291</a> ,<br><a href="#">292</a> , <a href="#">293</a> , <a href="#">294</a> , <a href="#">295</a> , <a href="#">296</a> , <a href="#">297</a> , <a href="#">298</a> , <a href="#">299</a> , <a href="#">300</a> , <a href="#">301</a> , <a href="#">302</a> , <a href="#">303</a> , <a href="#">304</a> , <a href="#">305</a> , <a href="#">306</a> , <a href="#">307</a> , <a href="#">308</a> ,<br><a href="#">309</a> , <a href="#">310</a> , <a href="#">311</a> , <a href="#">312</a> , <a href="#">313</a> , <a href="#">314</a> , <a href="#">315</a> , <a href="#">316</a> , <a href="#">317</a> , <a href="#">318</a> , <a href="#">319</a> , <a href="#">320</a> , <a href="#">321</a> , <a href="#">322</a> , <a href="#">323</a> , <a href="#">324</a> , <a href="#">325</a> ,<br><a href="#">326</a> , <a href="#">327</a> , <a href="#">328</a> , <a href="#">329</a> , <a href="#">330</a> , <a href="#">331</a> , <a href="#">332</a> , <a href="#">333</a> , <a href="#">336</a> , <a href="#">337</a> , <a href="#">338</a> , <a href="#">339</a> , <a href="#">342</a> , <a href="#">343</a> , <a href="#">344</a> , <a href="#">345</a> , <a href="#">348</a> ,<br><a href="#">349</a> , <a href="#">350</a> , <a href="#">351</a> , <a href="#">354</a> , <a href="#">355</a> , <a href="#">356</a> , <a href="#">357</a> , <a href="#">360</a> , <a href="#">361</a> , <a href="#">362</a> , <a href="#">363</a> , <a href="#">366</a> , <a href="#">367</a> , <a href="#">368</a> , <a href="#">369</a> , <a href="#">372</a> , <a href="#">373</a> ,<br><a href="#">374</a> , <a href="#">375</a> , <a href="#">378</a> , <a href="#">379</a> , <a href="#">380</a> , <a href="#">381</a> , <a href="#">382</a> , <a href="#">383</a> , <a href="#">384</a> , <a href="#">385</a> , <a href="#">386</a> , <a href="#">387</a> , <a href="#">388</a> , <a href="#">389</a> , <a href="#">392</a> , <a href="#">393</a> , <a href="#">394</a> ,<br><a href="#">395</a> , <a href="#">396</a> , <a href="#">397</a> , <a href="#">398</a> , <a href="#">399</a> , <a href="#">400</a> , <a href="#">401</a> , <a href="#">402</a> , <a href="#">403</a> , <a href="#">404</a> , <a href="#">405</a> , <a href="#">406</a> , <a href="#">407</a> , <a href="#">410</a> , <a href="#">411</a> , <a href="#">412</a> , <a href="#">413</a> ,<br><a href="#">416</a> , <a href="#">417</a> , <a href="#">418</a> , <a href="#">419</a> , <a href="#">420</a> , <a href="#">421</a> , <a href="#">422</a> , <a href="#">423</a> , <a href="#">424</a> , <a href="#">425</a> , <a href="#">426</a> , <a href="#">427</a> , <a href="#">430</a> , <a href="#">431</a> , <a href="#">432</a> , <a href="#">433</a> , <a href="#">436</a> ,<br><a href="#">437</a> , <a href="#">438</a> |
| \GTS@secondColor .....                    | 11, <a href="#">28</a> , <a href="#">29</a> , <a href="#">42</a> , <a href="#">46</a> , <a href="#">54</a> , <a href="#">58</a> , <a href="#">189</a> , <a href="#">193</a> , <a href="#">197</a> , <a href="#">207</a> , <a href="#">213</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| \GTS@secondColorBackup .....              | 13, <a href="#">28</a> , <a href="#">86</a> , <a href="#">229</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| \GTS@secondColorTemp .....                | 27, <a href="#">29</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSa .....                               | 4, <a href="#">138</a> , <a href="#">141</a> , <a href="#">142</a> , <a href="#">143</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| \GTSadj .....                             | 4, <a href="#">138</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSadjacency .....                       | 4, <a href="#">138</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSadjacencyMonochrome .....             | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| \GTSadjacencyMonochromeRelation .....     | 7, <a href="#">320</a> , <a href="#">321</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| \GTSadjacencyRelation .....               | 4, <a href="#">141</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSadjMon .....                          | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| \GTSadjMono .....                         | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| \GTSadjMonoRel .....                      | 7, <a href="#">320</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSadjMonRel .....                       | 7, <a href="#">320</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSadjRel .....                          | 4, <a href="#">141</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSaM .....                              | 7, <a href="#">321</a> , <a href="#">322</a> , <a href="#">323</a> , <a href="#">324</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| \GTSaMR .....                             | 7, <a href="#">320</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSaR .....                              | 4, <a href="#">141</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSdAD .....                             | 5, <a href="#">167</a> , <a href="#">168</a> , <a href="#">169</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| \GTSdADM .....                            | 8, <a href="#">356</a> , <a href="#">364</a> , <a href="#">366</a> , <a href="#">367</a> , <a href="#">368</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| \GTSdADMR .....                           | 8, <a href="#">362</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSdADR .....                            | 5, <a href="#">166</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSdAU .....                             | 5, <a href="#">150</a> , <a href="#">153</a> , <a href="#">154</a> , <a href="#">155</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| \GTSdAUM .....                            | 7, <a href="#">332</a> , <a href="#">340</a> , <a href="#">342</a> , <a href="#">343</a> , <a href="#">344</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| \GTSdAUMR .....                           | 7, <a href="#">338</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSdAUR .....                            | 5, <a href="#">152</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| \GTSdirAdjDown .....                      | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| \GTSdirAdjDownMono .....                  | 8, <a href="#">356</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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